

Leiry Hiciano, Jasmine Emeruwa, Michaela James, Michelle Akem, Nabeel Haider, Joshua Chavarria
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Group 4

Project Final Report

Introduction:

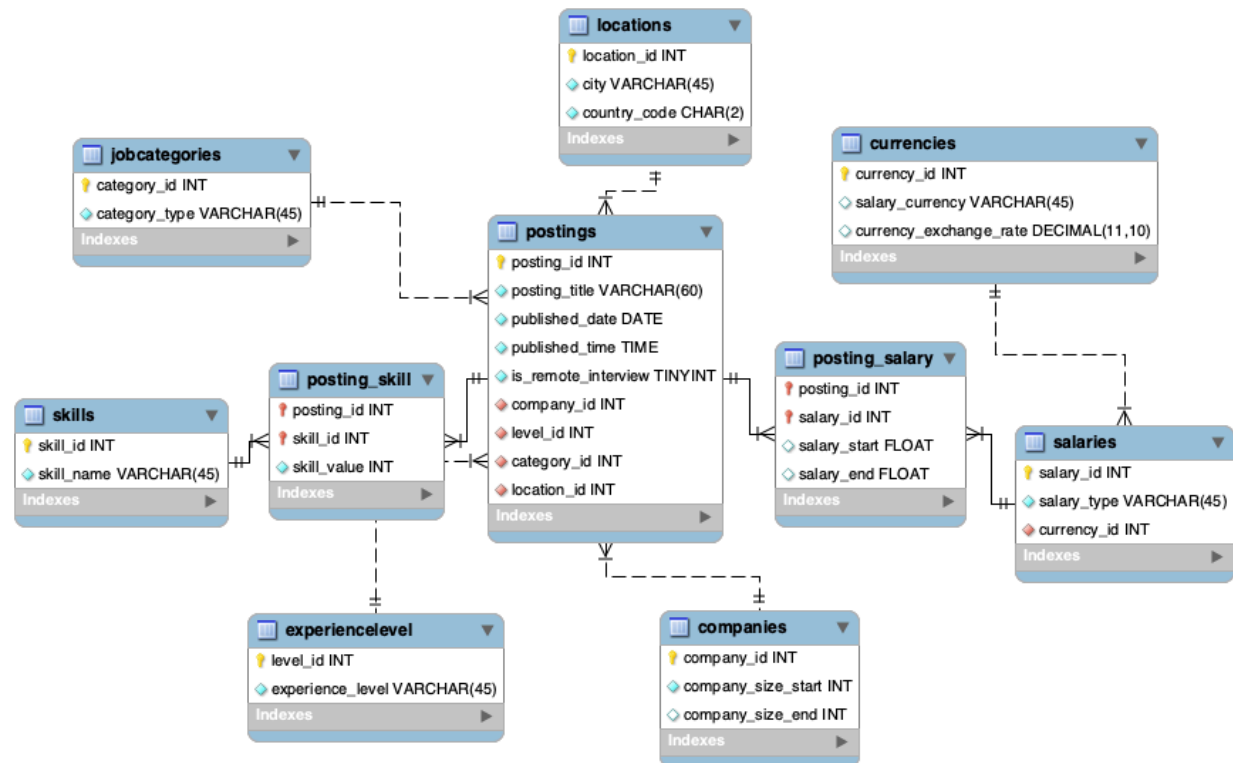
In today's world, IT professionals find themselves getting lost or not knowing where to look because everything is constantly changing in this area of work and different jobs require a certain combination of skills and knowledge sets. Our database stands out as a source of effectiveness and knowledge in the crowded field of IT recruitment because it recognizes the need for a more efficient approach to job finding. Additionally, our database also functions as an incredible platform for trend forecasting, industry analysis, and a repository of job postings. The users can find different trends in things like income distribution, locate the most in-demand skills, and understand the changing IT job market environment with the help of insightful analytics. The database is more than just a tool because it can also help them find interesting job opportunities and give them the clarity and confidence needed to face the challenges of the modern job market effectively.

Database Description:

Our database aims to serve as a platform for IT professionals seeking job opportunities in various locations and with diverse skill sets and experience levels. It will include filters such as Job Postings, Companies, Skills, and Experience Levels but be limited to a certain number of months to reduce the amount of data. The database will be able to answer questions about the highest average salary, the most sought-after programming languages, and more. Our database will include the JOB postings which represent individual job listings, each associated with specific attributes such as job title, company, location, required skills, and experience level. The companies inform people of the details about the organizations offering job opportunities. info such as company size provides insights into potential employers for job seekers. Some of the

skills are HTML, CSS, JavaScript, etc. Lastly is experience level, which categorizes job postings based on the experience required, ranging from entry-level to senior positions.

Logical Design:



Entities:

- salaries: has information on salary type and currency
- currencies: has information about different currencies and their exchange rates
- locations: shows cities and countries for all the postings
- skills: has the skill names for the job postings
- jobcategories: categorizes different types of jobs, like remote, hybrid, or in-person
- experiencelevel: represents levels of experience, such as entry-level, mid-level, or senior
- postings: main entity showing job postings and their corresponding ties to other information
- companies: shows company size information
- posting_salary: holds information about salary ranges

- posting_skills: holds information about skill values for specific jobs

Our main reason behind structuring our database this way was to easily update information that didn't depend on a specific posting, such as experience level or currency, which would allow us to provide the most accurate data. With postings as our central table, each row represents a unique posting with information that only depends on the posting ID.

Physical Database:

- The physical database structure was forward-engineered from our ERD. The data was imported from CSV created in Google Sheets with manual ID input, and the database was backed up and exported in an SQL file dump for team use.

Sample Data:

- From the original dataset, we used Pandas and Google Sheets to truncate our data. First, using Python and regex, we created new columns for the published date and time extracted from the original "Published_at" column. Next, to match our data to our ERD, we formatted the salary information to provide a salary type, salary start, and salary end for each row, including the ones with more than one type. Then, to simplify our imported data, we dropped the multiple currency columns because they were the same in each row, filtered for postings with only one salary type, and kept data that was through 10-01 to 11-15 and had company sizes that ranged instead of single values. Finally, we dropped rows such as Marker_icon and Open_to_hire_Ukrainians and took the last 150 rows as our sample data.

Views/Queries:

Query	Explanation	Req. A	Req. B	Req. C	Req. D	Req. E
locations_offering_most_jobs	joins the locations and postings tables to count the number of job postings for each location, allowing us to see which locations have the highest number of job postings	☒	☐	☒	☐	☐

preferred_workplace_for_junior_roles	selects the posting IDs for junior-level roles (level_id = 3) and then joins with the postings and job categories tables to count the number of these junior-level postings for each category type (e.g., "Remote", "Hybrid", "Onsite") shows the preferred workplace types or arrangements for junior-level job roles	☒	☒	☒	☐	☒
high_earning_jobs	calculates the median salary for each job posting by subtracting the start and end salaries, and then compares it to the overall average median salary across all postings helps identify high-earning job opportunities based on the salary range offered	☒	☒	☒	☒	☒
skills_by_experience_level	groups skills by their popularity (count of postings) for each experience level shows the most popular skills required for different levels of experience, providing insights into the skills emphasized for entry-level, mid-level, or senior roles	☒	☒	☒	☒	☐
skill_average	calculates the average skill value for each skill, considering only skills that appear in at least five job postings and have an average value of 3 or higher identifies the most highly valued skills across job postings	☒	☒	☒	☐	☐

company_sizes	retrieves data from the postings table within the it_postings database and calculates the number of postings associated with each unique company. It categorizes companies into size groups based on their posting count: those with 10 or fewer postings are labeled as "Small Company", those with more than 10 but fewer than or equal to 50 postings are labeled as "Medium Company", and those with more than 50 postings are labeled as "Large Company"	☐	☒	☒	☐	☐
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Change from Our Initial Design:

From our initial design, we made many changes considering the lessons we learned about data ethics and tailoring our database for our audience. The original dataset was a broad view of IT job postings for all of Poland, but because we wanted our database to be a resource for college students, it made us more aware of what columns would be unnecessary for this audience. For instance, the Ukrainian hiring column wasn't entirely relevant to most college students, so it was excluded from the final database. Additionally, one of the hardest aspects of this project was cutting down the dataset and the data cleaning. It had to be executed in a way that didn't diminish the diversity of job postings. This is why we even had to strategically omit data from the top of the database instead of the bottom because the more recent data has many senior roles that aren't attainable to college students. As for data cleaning, some of the measures taken to ensure our data was easy to consume were dividing the publishing stamp into dates and times, only including postings with one salary type, and removing odd characters from the posting titles. Overall, we tried our best to make the necessary changes per our team mentors' recommendations.

Database Ethics Considerations:

The database does seem likely to be an inclusive information resource in its domain, as we include job listings from different places and for people from all backgrounds. Though most of

our listings are from Poland, we do include other locations such as postings from Great Britain and the Netherlands. This allows us to show the diversity of job options in the IT industry. We are also inclusive of the types of skills an applicant should have, the positions that are listed, as well as their experience level. There are companies of various sizes included, ensuring that smaller start-ups with even 20 people have a chance to get applicants, such as the bigger companies with hundreds of employees do.

However, there are some inherent biases within the database. Primarily, the positions are certainly leaning toward Polish IT positions than those of other countries. The majority of the workplace type is remote and a good portion of the posted experience levels are mid to senior level, which cuts out a lot of applicants – especially if they are young or new graduates. These partialities to a certain side do pose a threat to the inclusivity of our database, as biases can deter users.

To avoid bias and integrate inclusiveness, my team will have to ensure that we vary the IT positions we include. For instance, we should add a lot of positions that are not just calling for mid to senior-level applicants, or workplaces that are not just remote. This will ensure that applicants from diverse backgrounds and with varying experience levels have equal opportunities to access and benefit from the database. By reducing entry barriers and placing various opportunities in view of applicants, we create a more inclusive platform that accommodates the needs and preferences of a broader spectrum of job seekers.

Additionally, ethics should be considered when creating this database, since things like the transparency of data sources, the provision of accurate and impartial information, and the protection of user privacy are relevant, even with the more collective data we use.

Lessons Learned:

I think that throughout the whole project, we had many lessons and we learned a lot, not only about our database and fixing our ERD or our queries but also lessons about working with others. One crucial takeaway was recognizing each team member's strengths early on, and ensuring tasks were delegated accordingly. This will lead to the best work being submitted for

the whole team because everyone puts in their best work on something they are good at. For instance, some of our group members were good at writing queries while others felt more comfortable with data analysis, so we assigned those respective tasks for the final project. Now some of the lessons that we learned when working on the database was that using precise data types for the situation is key. For example, our salary information initially used FLOAT data types, but this made it harder to format the sets when analyzing the high-earning jobs through our query. Therefore, we had to change the data type to DECIMAL, and this was the better choice as it allowed us to round our dollars properly. The most important lesson was the importance of communication. We encountered an issue where individuals were working on the same task for our Database Design submission, which was bad because one of the key elements, the sample data, was left unfinished. To resolve this issue, we worked together to complete the section at the last minute and for subsequent projects emphasized the task assignments. One lesson we'll take away from this is to start assignments earlier and provide multiple means of communication.

Potential Future Work:

If we were to continue working on this database, one improvement would be to delete duplicate job postings that have everything similar except for the location. This was expected for our database because larger companies can often hire for the same role in different areas, but it introduced inherent repetitiveness that diminishes our data quality. One way to fix this would be to manually add a "Multiple Locations" row set to a BOOLEAN data type. We weren't allowed to modify the data for this assignment, but adding this later would be helpful to signal to our audience that there's a wider range of opportunities for one job and maybe more competition.

In addition to this, we would also suggest adding a company name column to our companies table, which would increase the information provided to the audience because right now the uniqueness of the entries is only based on the company size. Two companies can be entirely different but have the same size, so this omits information from the user. This information wasn't included in the dataset, but it would be extremely helpful if college students had certain companies in mind that they would want to work for.

Citations

Job Offers IT - Just Join IT. (n.d.). Just Join IT. Retrieved May 10, 2024, from <https://justjoin.it/>

RSKriegs. (2022). *Polish IT job board data from 2022*. Kaggle.com.

<https://www.kaggle.com/datasets/kriegsmaschine/polish-it-job-board-data-from-2022>